The Greedy Digital-Root Sequence Is a Permutation

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Abstract

We prove the permutation conjecture for OEIS A340138. The argument uses the finite set of digital roots: any root occurring infinitely often forces every other root to do so, and then every positive integer must appear. The same proof applies in every integer base at least two. We explicitly resolve an indexing inconsistency in the entry by following its published terms and Mathematica program.

1 The precise sequence

For an integer base $b \geq 2$, let $\mathrm{dr}_b(0) = 0$, and for $x > 0$ let

$$\mathrm{dr}_b(x) = 1 + ((x-1) \bmod (b-1)) \in \{1, \ldots, b-1\}.$$

This is the usual iterated digit sum. Start with $a_b(1) = 0$ and, for $n \geq 1$, take $a_b(n+1)$ to be the least unused nonnegative integer containing the digit $\mathrm{dr}_b(a_b(n))$ in its ordinary base-b expansion. Thus $a_b(2) = b$.

At $b = 10$ these are exactly the displayed terms and the Mathematica program in OEIS A340138 [1]. The entry's prose instead puts $a(n-1)$ in the rule for $a(n+1)$ while specifying only $a(1)$; that rule is not initialized and disagrees with its program. We use the immediate-predecessor indexing above, as established by both the program and data. The permutation assertion below concerns that published sequence; it does not silently supply an unspecified initial value for a different two-step rule.

Theorem 1. *For every $b \geq 2$, the sequence $(a_b(n))_{n \geq 1}$ is a permutation of the nonnegative integers. In particular the permutation conjecture in A340138 is true for its published sequence.*

Proof. Every stage is well-defined: infinitely many base-b integers contain any specified digit, while only finitely many have been used. Terms are distinct by construction, and every term after the initial zero is positive.

For $1 \leq r < b$, let $\mathcal{C}_r$ be the positive integers containing the digit r. If the digital root r occurs at infinitely many terms, then every $M \in \mathcal{C}_r$ must appear. Indeed, if M stayed unused, every step immediately following a term of root r would choose an unused member at most M. The infinitely many such choices are distinct, contradicting the finiteness of $\{1, \ldots, M\}$.

At least one positive root r occurs infinitely often, since the set of possible positive roots is finite. Fix any other root $s \in \{1, \ldots, b-1\}$, and choose the unique $d \in \{0, \ldots, b-2\}$ satisfying $r + d \equiv s \pmod{b-1}$. For every $L \geq 1$, the base-b word

$$r \underbrace{0 \cdots 0}_{L \text{ zeros}} d$$

represents a distinct positive member of $\mathcal{C}_r$ with digital root s. The preceding paragraph forces all these integers to appear. Therefore s also occurs infinitely often. This proves that every positive root recurs infinitely often.

Finally every positive integer contains at least one nonzero digit s, so it belongs to $\mathcal{C}_s$ and must appear. The initial zero already appears exactly once. This proves the permutation assertion. □

2 Independent verification and scope

The accompanying `verify.py` compares all 10,000 published decimal terms with an exact implementation. A separate direct candidate scan, using repeated digit sums rather than the modular digital-root formula, verifies the first 200 terms in bases 2 through 16. The constructive root-propagation words are checked for every pair of roots in those bases and lengths 1 through 20. These computations validate the implementation and the intended indexing; the proof establishes the entire infinite permutation. No finite computation is used to infer surjectivity.

References

[1] M. De Vlieger, OEIS A340138, https://oeis.org/A340138, accessed October 26, 2026.